Create a cute AI generated picture of web accessibility for my HCI class [ChatGPT]

CSE472: HCI – Human Computer Interaction

Lecture – 16: Web Accessibility

Spring 2026

BRAC UNIVERSITY

Inspiring Excellence

Web Accessibility

- Web accessibility refers to the practice of removing barriers that prevent interaction with, or access to, websites by people with disabilities so that all users have equal access to information and functionality.

- Web content is extremely visual, so people with vision impairments are particularly affected
  - Web developers need to especially pay attention to needs of visually-impaired people, e.g., blind, low-vision, color blind, etc.

Why Make Things Accessible?

- Good for business
  - Reach a large audience.
- Support social inclusion
  - Participation from a diverse group is good.
- Follow the law
  - Access to information is a basic human right.

Legal Support for Accessibility

- 2009: Right to Information Act (Bangladesh)
- 2013: Rights and Protection of Persons with Disabilities Act (Bangladesh)
- 2016: Rights of Persons with Disabilities Act (India)

Source: https://legislativediv.portal.gov.bd/

https://www.refworld.org/legal/legislation/natlegbod/2013/en/148293

https://www.indiacode.nic.in/

Legal Cases

- 2000, Maguire v. Sydney Organizing Committee for the Olympic Games (Australia)
  - Olympic Games website was unusable for blind users.
  - Tribunal treated the website as a public service.
  - Website had to be updated so blind users could access the same information.
- 2009, National Human Rights Commission of Korea (Web Accessibility Complaint)
  - Public websites were inaccessible to visually impaired users
  - Users with visual disabilities took 3 minutes 49 seconds (3.2× longer) to complete tasks.
  - Public institutions were asked to improve websites to reduce this time gap.
- 2025, Vaishnavi Jayakumar v. Election Commission of India
  - ECI website could not be accessed by users with disabilities.
  - Court treated the website as essential election information.
  - ECI was directed to take steps to make the website accessible.

What are Accessibility Features?

They are adjustments or built-in tools that remove barriers for people with:

- *Visual impairments*
- *Hearing impairments*
- *Mobility limitations*
- *Speech difficulties*
- *Cognitive or learning disabilities*

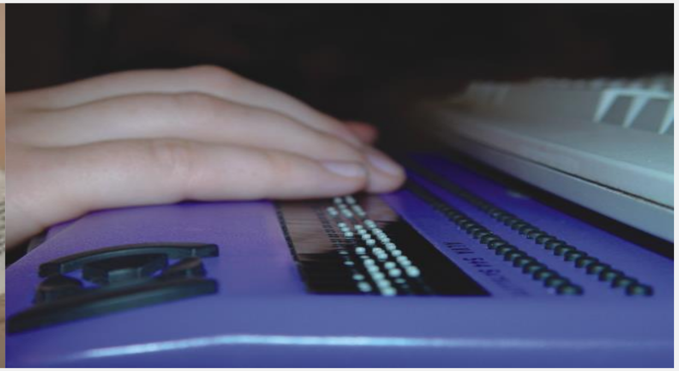
Accessibility Features / Technologies Improve Experience and Quality of Life

- Enables a person to function at his or her own pace.
- Fosters independent living
- Maintains or improves daily function
- Reduces stress-related injuries
- Aids integration into society (levels the "playing field")
- Modifies the environment instead of the person

Accessibility Features/Technologies Can Be Simple or Complex

- A magnifying glass
- A straw
- Anti-glare screen for the monitor
- Door handles instead of door knobs
- Calculators/clocks with large digits

Source: https://vaalio.co.za/diane-schoeman

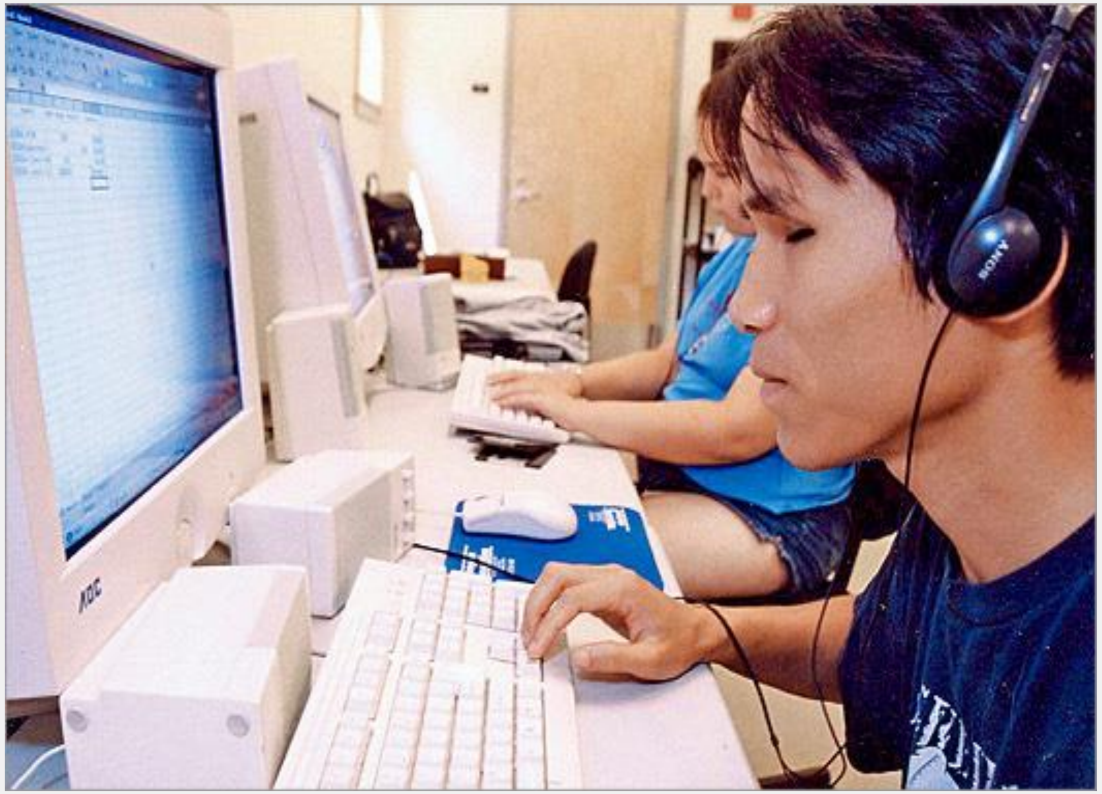


Source: https://www.cdb.org.il/en/communication-methods/braille/

- Alternative keyboards
- Braille and refreshable braille
- Scanning software
- Screen magnifiers
- Screen readers
- Speech recognition
- Speech synthesis
- Tabbing through structural elements
- Text browsers / Voice browsers
- Visual notification

Web Accessibility for Blind or Visually Impaired People

- Need non-visual access to screens



Speech / Screen reader

Source: https://clarkwp.wordpress.com/2015/11/23/web-accessibility-screen-magnification/

Refreshable Braille Display

Source: https://www.kimbervie.nl/computeraanpassingen.htm

Web Accessibility for Deaf or Hard of Hearing People

- Need visual access to content

Sign Language

Captioning

Source: https://commons.wikimedia.org/wiki/File:TV_Interpreter.jpg

Source: https://www.nytimes.com/2018/02/19/technology/personaltech/closed-caption-content-online.html

Web Accessibility for People with Mobility Impairments

Duchenne muscular dystrophy (DMD) in a person using a motorized wheelchair and breathing ventilator

The orbiTouch creates a keystroke when you slide the two domes into one of their eight respective positions

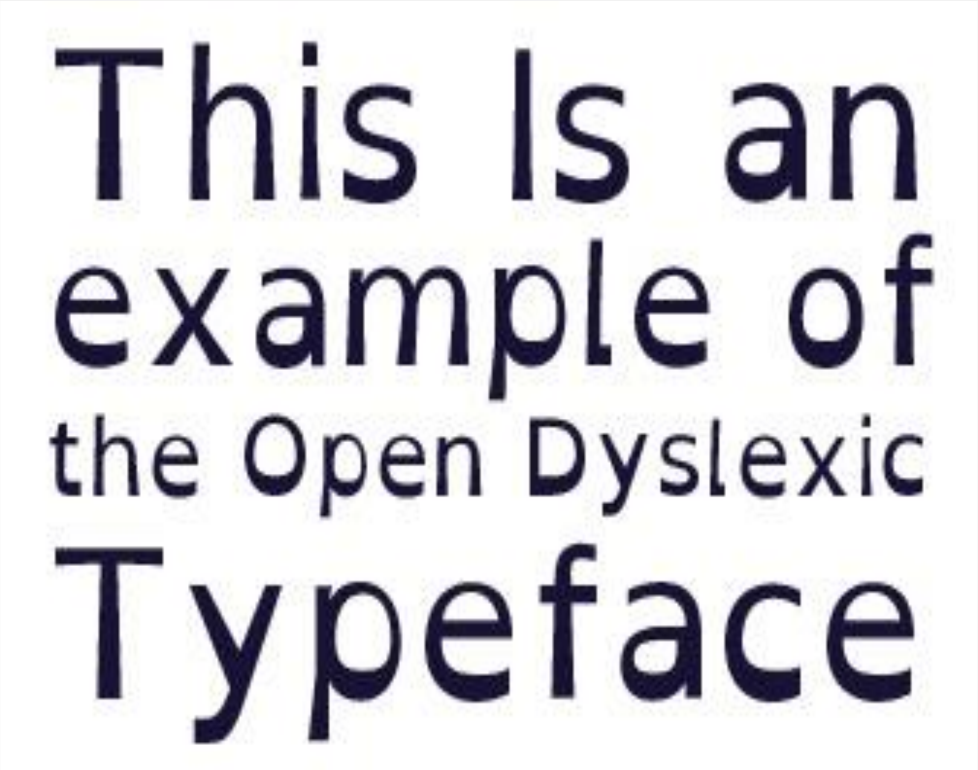
Foot mouse

DAISY Victor Reader

aimed for people who get pain in their hands from typing with a traditional keyboard

Web Accessibility for People with Cognitive Impairments

Memory
Attention
Literacy
Learning
and more…



Source: https://medium.com/codex/accessibility-fonts-and-dyslexia-3cc495795127

Web Accessibility Principles

- Make things:
  - Perceivable - Users must be able to **see or hear** the content
  - Operable - Users must be able to **use and control** the website
  - Understandable - Users must be able to **understand** the content and how the site works
  - Robust – the website must work well with **different technologies**, including **assistive tools**

Web Content Accessibility Guidelines

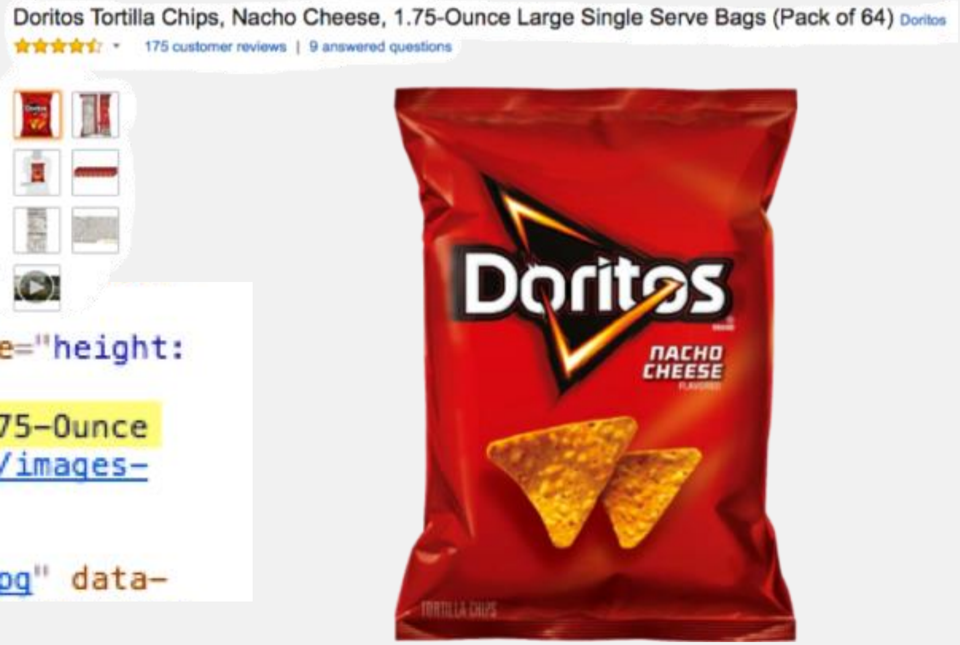
- Perceivable
  - Provide text alternatives for non-text content and provide captions and alternatives for audio and video content.
  - Make content adaptable; and make it available to assistive technologies.
  - Use sufficient contrast to make things easy to see and hear.
  - *Example: A blind user uses a screen reader. The screen reader reads the text and describes images so the user understands the page.*
- Operable
  - Help users find content and make everything keyboard accessible.
  - Give users enough time to read and use content.
  - Do not use content that causes seizures.
  - *Example: A person with limited hand movement uses only a keyboard to move through menus and click buttons.*

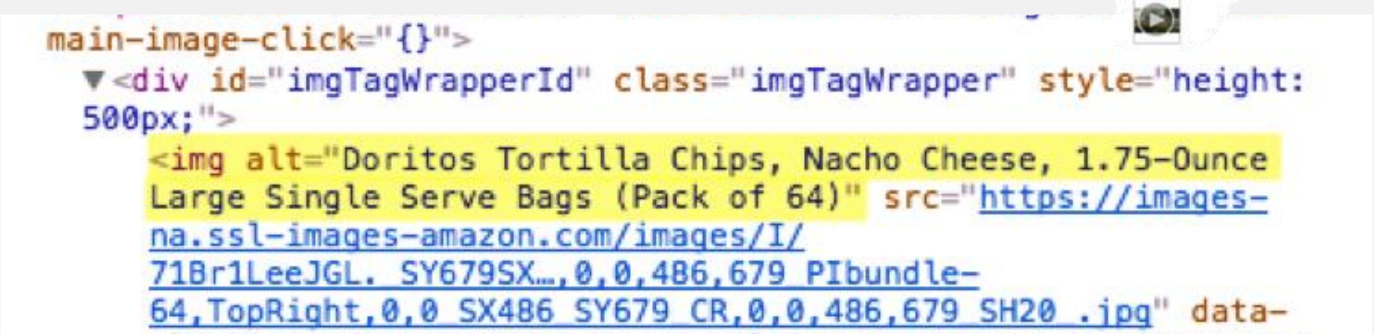
Web Content Accessibility Guidelines

- Understandable
  - Make text and content understandable and readable.
  - Make content operate in predictable ways and help users avoid and correct mistakes.
  - Follow and use standards.
  - *Example: The website layout and menu stay the **same on every page**.*
- Robust
  - Maximize compatibility with current and future technologies.
  - Doesn’t break every time there is an OS update.
  - Works across a variety of services and platforms.
  - *Example: A website works properly with a screen reader today and will still work when technology updates in the future.*

Practical Tips / Guidelines

- Provide appropriate alternative text
- Use the <alt> attribute to describe the function of any image or animation.





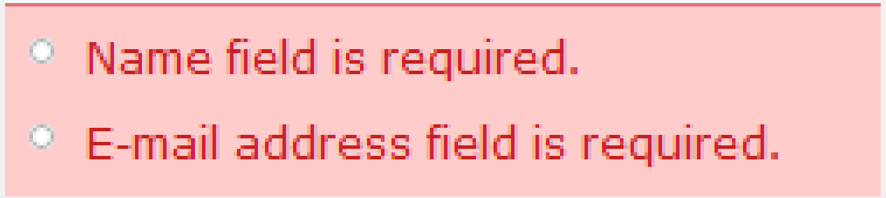
Accessibility Tips / Guidelines

- Provide appropriate document structure
  - Helps understandability
  - Helps for cognitive impairments
  - Helps for keyboard navigation / shortcuts
- Info on Page organization. Use headings, lists, and consistent structure. Use CSS for layout and style. Allow users to skip repetitive elements on the page.
- Provide headers for data tables
  - Helps screen reader users to navigate and understand the data table

Accessibility Tips / Guidelines

- Ensure links make sense out of context
  - Use text that makes sense when read out of context. For example, avoid "click here" or "more".
  - More guidelines for Hypertext links.

- Ensure users can complete and submit all forms
  - Every form element (text field, checkbox, etc.) has a label
  - Use the <label> element.
  - Make sure the user can submit the form and recover from any errors (e.g., failure to fill in all required fields).



Accessibility Tips / Guidelines

- Caption and/or provide transcripts for media.
- Guidelines for captioning and transcripts of audio, and descriptions of video.
- Scripts, applets, & plug-ins. Provide alternative content in case active features are inaccessible or unsupported.
- Ensure accessibility of non-HTML content, including PDF files, Microsoft Word documents, PowerPoint presentations and Adobe Flash content.

Source: https://techtrendske.co.ke/2016/02/11/no-more-flash-player-google-saysshift-to-html5/

Accessibility Tips / Guidelines

- Do not rely on color alone to convey meaning!
- Design to standards
    - HTML compliant and accessible pages are more robust and provide better search engine optimization.
    - CSS allows you to separate content from presentation.
    - More flexibility and accessibility of your content.

Verifying Accessibility

- Testing with real users is the best way to validate
- Build in accessibility as part of the design
  (NOT as an afterthought)
- Check your work. Validate.
- Online materials can help:
  - http://www.w3.org/TR/WCAG
  - http://webaim.org/intro/#principles
  - http://webaim.org/standards/wcag/checklist

Summary

- Web accessibility refers to the practice of removing barriers that prevent interaction with, or access to, websites by people with disabilities so that all users have equal access to information and functionality.

- There are well established guidelines and checklists for how to make your designs and systems accessible.
- Follow and use these guidelines in your design process!

- Do NOT sacrifice accessibility for aesthetic!

Any Questions?

Acknowledgment: Lecture content adapted from Cornell University HCI & Design (Prof. Nicki Dell)